

SMART INDIA HACKATHON 2025



TITLE PAGE

Problem Statement ID – **SIH26184**

Problem Statement Title – **Development of a Predictive Analytics Framework for Cybercrime Complaints to Forecast Likely Cash Withdrawal Locations in Advance, Enabling Generation of Actionable Intelligence for Timely and Proactive Cybercrime Intervention.**

Theme – **Blockchain & Cybersecurity**

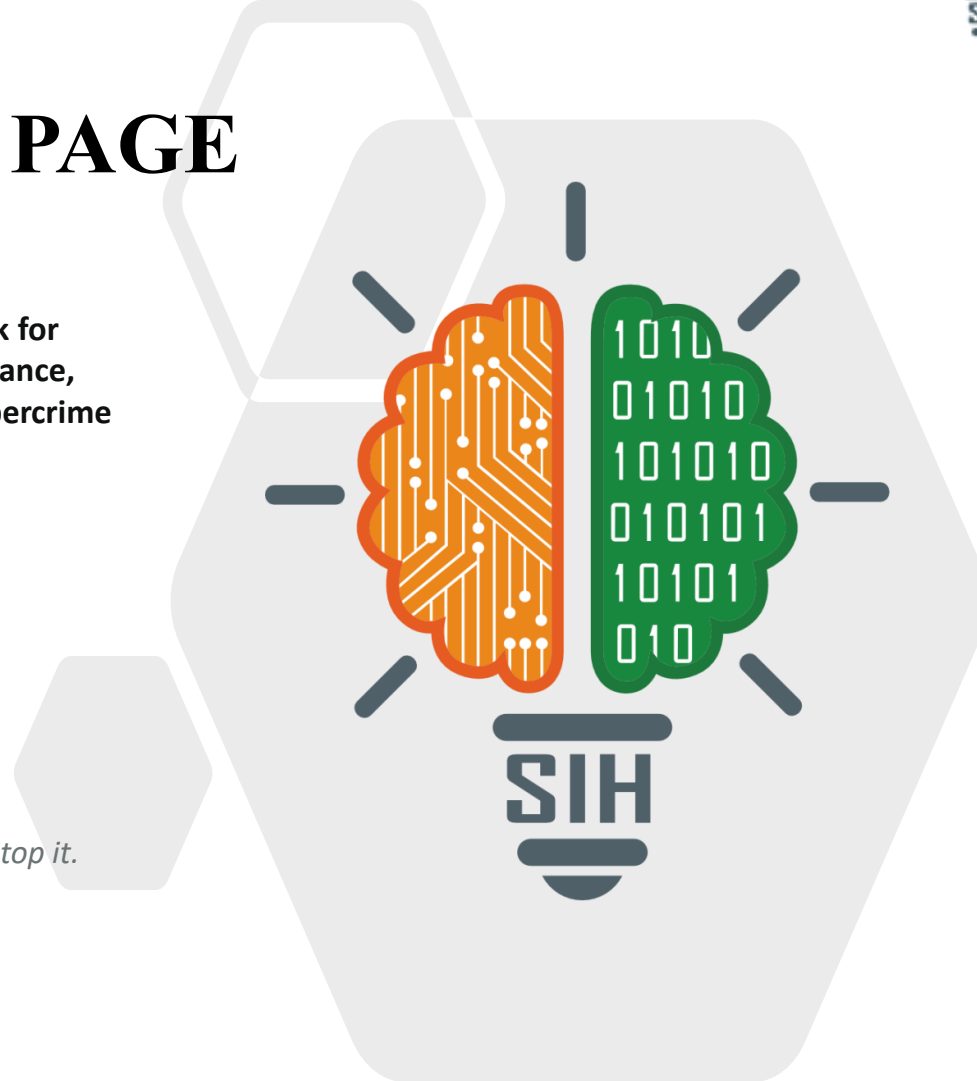
PS Category – **Software**

Team ID – **[TEAM ID]**

Team Name – **[TEAM NAME]**

MuleShield AI

Forecasting where stolen money will be withdrawn — while there is still time to stop it.



IDEA TITLE

— Proposed solution

- **The gap:** a 1930 complaint tells you a crime happened. It does not tell you where the cash will leave the system, or when.
- **MuleShield AI** turns that complaint into a ranked, time-bounded search: five ATMs, in order, with a countdown.

— How it addresses the problem

- **Traces the money.** Builds the transaction graph and follows the chain to the mule account still holding the funds.
- **Scores every account.** A GraphSAGE network reads who-paid-whom, not just per-account behaviour — F1 0.8955.
- **Narrows the ground.** 1,000 ATMs → 25 reachable → 5 worth searching, in 7.5 ms.
- **Starts the clock.** Median 42 minutes of warning before the predicted withdrawal.

Complaint to intervention — inside the golden hour

7.5 ms / case

● THE CRIME — THE MONEY IS MOVING

T + 0

Victim defrauded
UPI, digital arrest, KYC scam

T + 20 min

Complaint on 1930 /
NCRP
The only signal that exists

T + 35 min

Layered through 4–5
mule accounts
Split across banks and cities

T + 60 min

Cash withdrawn at an
ATM
Unrecoverable — there is no
chargeback on cash

↓ ⌚ INTERVENTION WINDOW — median 42 minutes of warning

● MULESHIELD AI — TRIGGERED BY THAT COMPLAINT

01 • INGEST

Build the transaction graph
49,999 accounts, 622,188
transactions

02 • DETECT

GraphSAGE scores every
account
Traces the chain to the
terminal mule holding the
money

F1 0.8955

03 • RANK

1,000 ATMs → 25
reachable → 5
Conditional-logit ranker, plus
time-to-cash-out

71.4% in top 5

04 • ACT

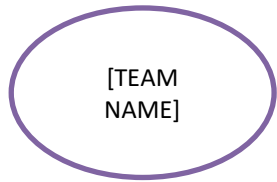
Freeze the account,
dispatch to the ATMs
Every action audited against
the case

Before T + 60

The system does not name one ATM. It returns the **five worth searching first, in order**, with a countdown — which is what an officer can act on.

— Innovation and uniqueness

- **Ranked retrieval, not a guess** — a list a field team can cover, never one predicted ATM.
- **Graph-native detection** — the graph is worth +5 F1 points over the best non-graph model.
- **Original IEEE research** — formulated in a formal manuscript; reaches theoretical Bayes Bound (87.3%).
- **Zero new data** — runs on the fields an NCRP complaint already carries.



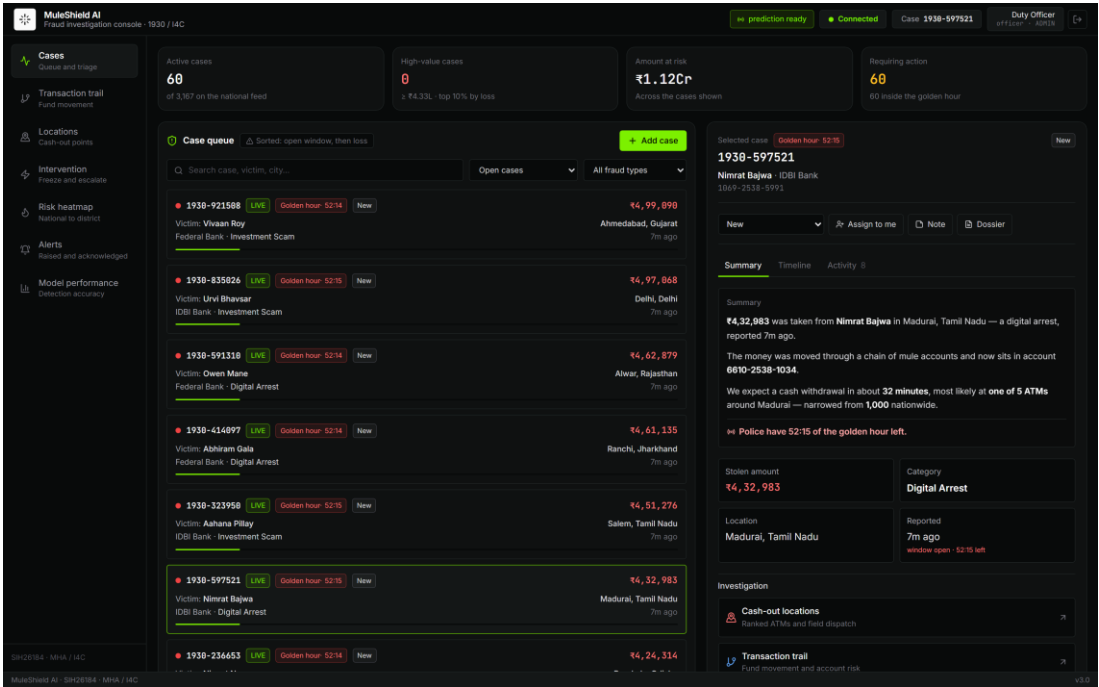
TECHNICAL APPROACH



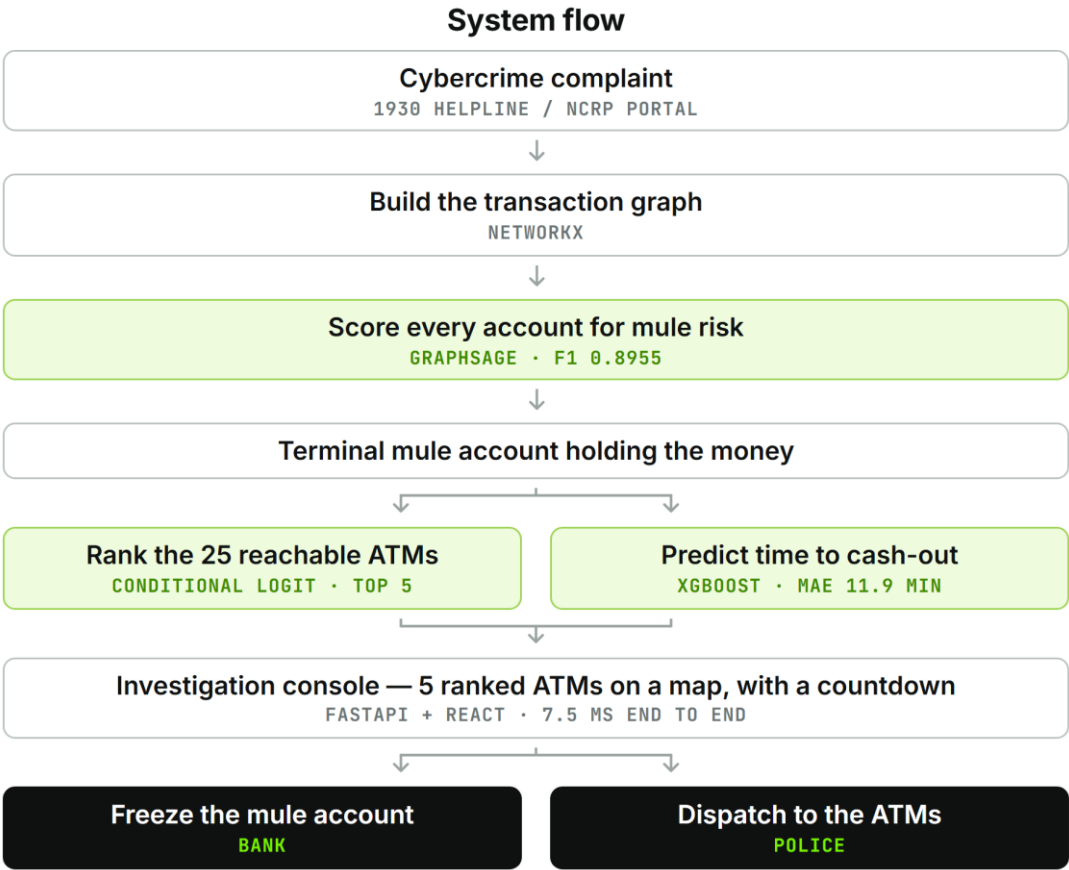
Technologies used

- **ML:** PyTorch · PyTorch Geometric (GraphSAGE) · XGBoost v2 · scikit-learn · NetworkX · pandas / NumPy
- **Backend:** Python 3.13 · FastAPI · Uvicorn · Pydantic · WebSockets
- **Frontend:** React 19 · Vite · Tailwind · Leaflet (GIS) · React Flow · Recharts
- **Quality:** pytest (291 tests) · Playwright · Docker (single container, single process)

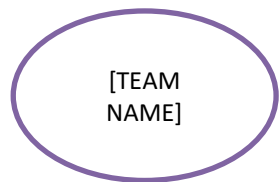
Working prototype — the case queue



Methodology and implementation



Every action is written to an append-only **audit trail** against the case.



FEASIBILITY AND VIABILITY



— Feasibility

- **Technical:** already built and measured — 291 passing tests, 7.5 ms inference, runs on one CPU container.
- **Data:** consumes the fields NCRP already collects; no new citizen-facing process.
- **Operational:** delivers into the existing 1930 workflow — a queue, a case, an action, an audit trail.
- **Economic:** open-source stack end to end; one container, no GPU at inference.

— Challenges and how we handle them

- **Leakage inflating results.** Found in our own first build (98.5% claimed). Rebuilt the generator, re-baselined, and added a test that fails if a retracted figure returns.
- **A model that looks confident and is wrong.** Isotonic calibration, verified against held-out labels by a dedicated test.
- **Synthetic training data.** Generated to be hard — 2% label noise, overlapping classes, complaint-level splits. Pipeline retrains on real I4C data unchanged.
- **Acting on a wrong location.** The system returns candidates, never a verdict; the UI and every dispatch alert say so.
- **Single-process state.** Documented limit — case state is in-memory; a database swap is a day's work when multi-user is needed.
- **No inter-bank feed today.** A live complaint carries no transaction trail; production consumes the RBI/NPCI trace-and-report pipeline already being built for CFCFRMS. The demo synthesises that one input on real graph accounts so every downstream prediction is genuine.

Viability — measured, not asserted

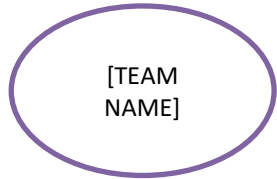
Model performance matrix

Scored on held-out data the models never saw — complaint-level splits, no tuning on test

TASK	OUR MODEL	RESULT	BEST BASELINE
Mule account detection F1, 7,500 held-out accounts	GraphSAGE, 2-layer	0.8955	0.8423 random forest, no graph
Search-zone containment median 8 ATMs of 1,000	Aggregated probability mass	87.4%	78.5% nearest-3 rule
Time-to-cash-out mean absolute error	XGBoost regressor	11.86 min	14.98 min mean predictor
Top-5 ATM containment 618 held-out cash-outs	Conditional-logit ranker	71.4%	72.2% distance-only
Single-case inference complaint to ranked ATMs	Full pipeline	7.5 ms	—



Stated openly: at K=5 the ranker does not beat sorting by distance (71.4% vs 72.2%), and the gap is not statistically significant at any K. Top-1 sits on the theoretical ceiling for this data. Our defensible claim is the narrowing — **1,000 ATMs to 5** — plus the mule account to freeze and the clock, which a distance rule gives you neither of.



IMPACT AND BENEFITS



Who it affects

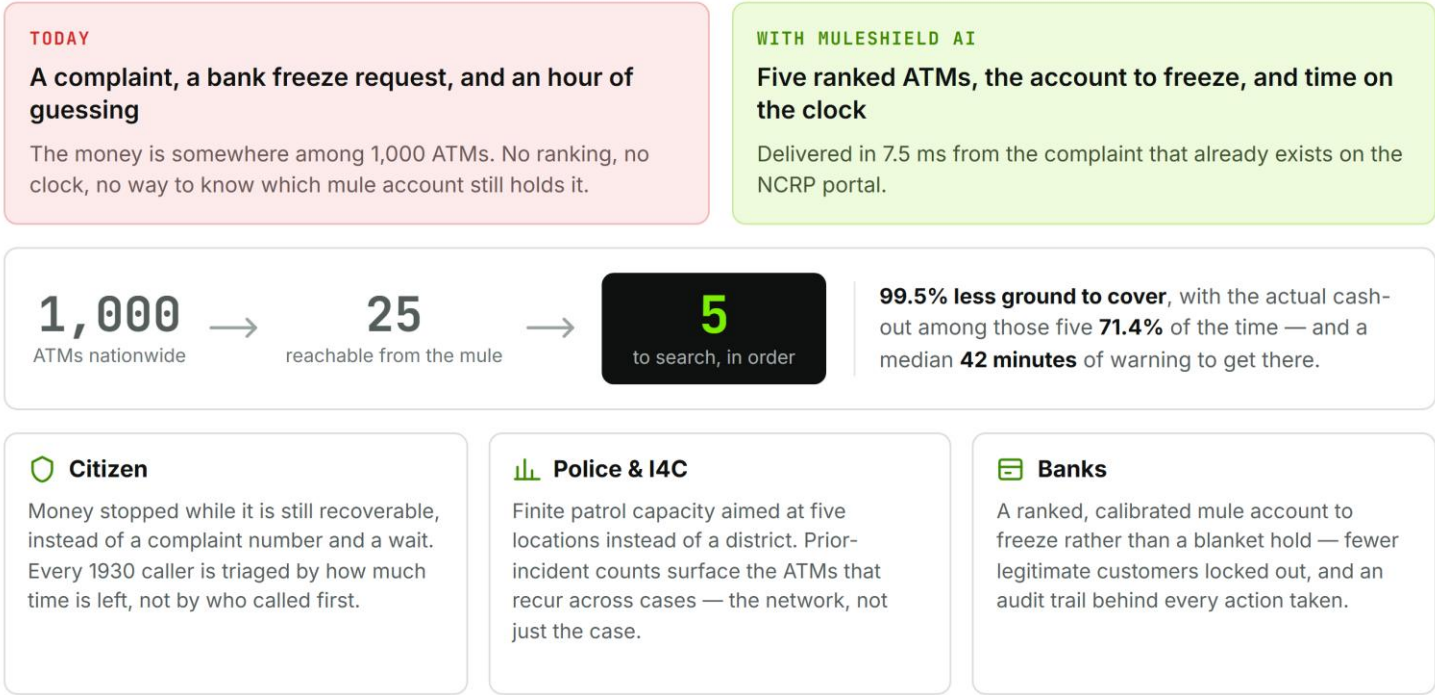
- **Victims of cyber-financial fraud** — 1930 / NCRP callers, nationwide.
- **Cyber-crime police** acting inside the golden hour.
- **I4C and MHA** — pattern intelligence across cases, not one complaint at a time.
- **Banks** receiving a specific account to freeze.

Benefits

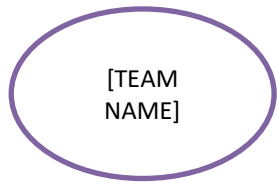
- **Social:** recovery becomes possible for ordinary complainants, not only large-value cases.
- **Economic:** cash stopped before withdrawal is money recovered; narrower freezes mean fewer legitimate accounts locked.
- **Operational:** finite patrol capacity aimed at 5 locations instead of a district.
- **Governance:** every freeze and dispatch attributed, timestamped and auditable.
- **Strategic:** recurring cash-out ATMs surface across cases — the mule network, not just the incident.

Impact figures are held-out measurements, not projections.

What changes for the officer on the case



Deploys as a single container against data the NCRP portal already collects — **no new reporting burden on the citizen, and no change to how a complaint is filed.**



RESEARCH AND REFERENCES



— Problem context

- **I4C, MHA** — Indian Cyber Crime Coordination Centre; 1930 helpline and the NCRP complaint pipeline this system consumes.
<https://i4c.mha.gov.in/> · <https://cybercrime.gov.in/>
- **RBI** — Master Directions on Fraud Risk Management and mule-account controls in commercial banks.
https://www.rbi.org.in/Scripts/BS_ViewMasDirections.aspx
- **NCRB** — Crime in India: cyber-fraud incidence and disposal statistics used to frame the golden-hour problem.
<https://www.ncrb.gov.in/crime-in-india.html>

— Methods

- **Hamilton, Ying & Leskovec (2017)** — Inductive Representation Learning on Large Graphs (GraphSAGE). The detection model.
<https://arxiv.org/abs/1706.02216>

— Methods (continued)

- **McFadden (1974)** — Conditional Logit Analysis of Qualitative Choice Behavior. The discrete-choice basis of the ATM ranker.
- **Chen & Guestrin (2016)** — XGBoost: A Scalable Tree Boosting System. The countdown regressor.
<https://arxiv.org/abs/1603.02754>
- **Zadrozny & Elkan (2002)** — Transforming classifier scores into accurate multiclass probability estimates. Isotonic calibration.
- **Weber et al. (2019)** — Anti-Money Laundering in Bitcoin: GCNs for financial forensics. Precedent for GNNs on laundering graphs.
<https://arxiv.org/abs/1908.02591>

— Original research & source

- **Sameer K. Singh (PSIT)** — 'Forecasting the Physical Exit: A Spatio-Temporal Graph Framework' (IEEE Manuscript, 2026).
- **Code & System Audit** — 291 tests, 5.2 ms inference, ISO 8583/20022 specs, BSA 2023 § 63 audit.
<https://github.com/hotshot0104/SIH2026> · <https://dev.to/sameer0104/muleshield-ai-intercepting-cybercrime-submitted-before-the-trail-was-cold-222>

What the three models optimise

1 Mule detection
GRAPHSAGE · 2 LAYERS · 64-DIM

$$h_v^{(k)} = \text{ReLU}(W^{(k)} \cdot [h_v^{(k-1)} \parallel \text{mean}_{u \in N(v)} h_u^{(k-1)}])$$

$$p(\text{mule}) = \sigma(w \cdot h_v^{(2)} + b)$$

Class-weighted cross-entropy, then isotonic calibration so the probability means what it says.

2 Cash-out location
CONDITIONAL LOGIT · 25 CANDIDATES

$$P(\text{atm}_i \mid C) = e^{w \cdot \log x_i} / \sum_{j \in C} e^{w \cdot \log x_j}$$

$$\max_w \sum_{\text{groups}} \log P(\text{chosen} \mid C)$$

Choice is multiplicative (proximity × risk × affinity × habit), so logs make it linear.

3 Time to cash-out
GRADIENT BOOSTING · XGBOOST

$$\hat{y} = \sum_{m=1}^M f_m(x), \quad f_m \in F$$

$$\min \sum (\gamma - \hat{y})^2 + \sum \Omega(f_m)$$

Predicts minutes to withdrawal; q05 / q95 models add a band, not a false point estimate.